
a Design and contrasts
16 adults, 45 biopsies; contrast colours carry through b to f
HR (n=8)
LR (n=8)
Training
Acute
HR -LR
Interaction
VL Biopsy | Proteomics
Ultrasound | mCSA
Leg Press | Strength
IHC | fCSA
T1
Training (T2-T1)
T2
Acute (T3-T2)
T3
Interactions (HR) - (LR)
b Proteins per contrast
1,900 tested; none at FDR < 0.05
HR
LR
T1
T2
T3
Training
Acute
100
50
0
50
100
proteins (down | up)
Down Π < 0.05
Up p < 0.05
Down p < 0.05
Up Π < 0.05
c Proteome
45 biopsies, 1,900 proteins
PERMANOVA
time R² 0.06, p = 0.0018
arm R² 0.07, p = 0.29
PC2 (9.1%)
PC1 (13.9%)
Phenotype
31 biopsies, six size and strength traits
PERMANOVA
time R² 0.04, p = 0.037
PC2 (21.4%)
PC1 (61.0%)
T1
T2
T3
HR
LR
d Pathways shared within each window
fgsea FDR < 0.05 in both arms of a window (interaction: either, top 5), shown in all nine contrasts; * opposite signs
Training
Acute
HR - LR
Interaction
Striated muscle contraction
Regulation of muscle contraction
Citric acid cycle TCA cycle
Proteasome: CLEC7A dectin 1 signaling
Mitochondrial gene expression
Eukaryotic translation elongation
Striated muscle contraction
mRNA splicing
Protein DNA complex organization
Sarcomere organization
MYC targets v1
Cardiac muscle tissue development
Regulation of muscle contraction
Aerobic respiration and respiratory elect.. *
Mitotic prometaphase *
Calcium signaling pathway *
Regulation of intracellular transport *
ATP synthesis coupled electron transport
Mitochondrial protein import
Mitochondrial membrane organization
Mitochondrial respiratory chain complex a...
Mitochondrial protein degradation
Citric acid cycle TCA cycle
Fatty acid beta oxidation
Mitotic spindle
Acyl CoA biosynthetic process
Cardiac muscle contraction
Glial cell differentiation
RHO GTPase effectors
Cytoskeleton dependent intracellular tran...
Complex iv assembly
Anchoring of the basal body to the plasma...
Mitochondrial biogenesis
Mitochondrial membrane organization
Cytoplasmic microtubule organization
HR
LR
HR
LR
T1
T2
T3
Training
Acute
NES
3
2
1
0
-1
-2
-3
-log10 FDR
2
6
12+
Database
Hallmark
Reactome
KEGG
GO BP
e Training signature rank
fry: each arm's p < 0.05 training sets in the other arm's ranking
LR training sets in HR's ranking
up (n = 73), 82% same way, FDR 0.016
down (n = 29), 76% same way, FDR 0.026
HR training sets in LR's ranking
up (n = 59), 90% same way, FDR 0.14
down (n = 46), 74% same way, FDR 0.25
0
500
1000
1500
rank in the other arm's training contrast, up to down (n = 1900)
f Acute signature rank
fry: each arm's p < 0.05 acute sets in the other arm's ranking
LR acute sets in HR's ranking
up (n = 29), 76% same way, FDR 0.0088
down (n = 44), 86% same way, FDR 0.0014
HR acute sets in LR's ranking
up (n = 101), 57% same way, FDR 0.44
down (n = 88), 77% same way, FDR 0.42
0
500
1000
1500
rank in the other arm's acute contrast, up to down (n = 1900)